

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

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ABSTRACT. Weak convergence of joint laws does not make disintegration continuous. We therefore keep a global weak belief state but prove Feller filtering on model-derived dominated Sobolev charts that control unnormalized kernels and their observation derivatives. Atomic microscopic observations enter by an exponentially good regularization. The unnormalized positive kernel, its evidence, and its projective zero-evidence direction form an exact Markov belief state with measurable dynamic programming. Tolerance-dependent bounded-Lipschitz coordinates approximate the full value, and the exact regular experiment is QMD, LAN, and locally Bernstein–von Mises.

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: EXACT DOMINATED
EXPERIMENTS, MEASURABLE BELIEF DYNAMICS, AND
TRIANGULAR-ARRAY INFERENCE

This section replaces the former Round-Seventeen source. The hidden-state transition and observation kernels are typed separately. Atomic, lattice, and continuous observations are treated on one stratified dominating measure; no artificial smoothing is used. Local asymptotic normality is proved as a uniform triangular-array likelihood expansion, and the Bernstein–von Mises theorem includes a separate testing and posterior-concentration argument.

1.1. The exact controlled experiment. Let E be the hidden state space and Y the observation space. For every admissible action $a \in A$ and parameter $\vartheta \in \Theta \subset \mathbb{R}^p$, let

$$M_{\vartheta}^a(x, dx')$$

be the hidden transition and

$$G_{\vartheta}^a(x', dy)$$

the observation kernel conditional on the new state. Their joint one-step kernel is

$$\Lambda_{\vartheta}^a(x, dx', dy) = M_{\vartheta}^a(x, dx') G_{\vartheta}^a(x', dy).$$

For the Sinai model, M is the exact A3 history transition; for hard spheres, it is the finite-volume law transition and its B4 kinetic limit. The observation coordinates are finite collections of the lattice marks, continuous roofs/coarea coordinates, and bounded path/contact cylinders.

Write Y as the disjoint union of finitely many strata Y_{α} , each of which is a product of a countable lattice and a Euclidean manifold. Let ν_{α} be counting measure times smooth volume on that stratum and set

$$\nu = \sum_{\alpha} 2^{-|\alpha|} \nu_{\alpha}.$$

On every reachable compact state/control/parameter chart, the A2 and B1 Fourier/coarea theorems give a density

$$G_{\vartheta}^a(x', dy) = g_{\vartheta}^a(x', y) \nu(dy)$$

with its atomic masses represented as densities on zero-dimensional strata.

Fix a reference hidden probability m_0 with positive density on every reachable regular component. A belief belongs to the regular chart $\mathcal{K}$ when

$$\pi(dx) = r(x) m_0(dx), \quad c \leq r \leq C, \quad \|r\|_{C_W^1} \leq C,$$

and the transition has density $m_{\vartheta}^a(x, x')$ with respect to $m_0(dx')$ satisfying the analogous bounds. These are verified below for the forward reachable set from the prepared initial family; they are not imposed on all of $\mathcal{P}(E)$.

Theorem 1.1 (Model-derived dominated regular chart). *For every finite horizon, compact parameter set, and compact relaxed-control class, the beliefs*

reachable from the A3 or B1–B4 prepared initial family lie in a forward-invariant set $\mathcal{K}$ on which:

- (i) M_ϑ^a and G_ϑ^a have the common dominating measures m_0 and ν above;
- (ii) their densities are bounded above and below on each reachable stratum;
- (iii) for some $s > \dim Y_\alpha + 2$ on every stratum,

$$\|m_\vartheta^a\|_{C_W^1} + \|g_\vartheta^a\|_{W_y^{s,1}(C_{W,x'}^1)} \leq C;$$

- (iv) the same bounds hold for three parameter derivatives and two control derivatives;
- (v) $\mathcal{K}$ is compact in weighted $L^1(m_0)$ and the transition maps it continuously into itself.

Proof. For A3, the reference density is the normalized inverse Jacobian of the exact history disintegration. Exponential stable contraction and the A2 weighted branch theorem give the C_W^1 state bound and a positive lower bound on each finite reachable regular chart. Fourier inversion with the A2 $(1+|b|)^{-2}$ tail may be repeated after inserting polynomial observation weights; the full finite-dimensional observation is partitioned into charts so that a fixed number of coarea variables yields s derivatives. Lattice and atomic coordinates are zero-dimensional strata and require no fictitious Lebesgue density.

For hard spheres, B1's fixed-size full-rank blocks give uniform $W^{s,1}$ densities in all continuous observation directions, including position constraints; B2's positive controlled evolution and B4's transfer lemma give the common state domination on every prepared finite-horizon reachable set. Source derivatives in A2/B1/B2 are exactly derivatives of the finite-volume likelihood kernel before normalization, and their global Fourier majorants justify differentiation in y and in ϑ under the inverse transform.

Bayes multiplication and prediction preserve positive lower and upper bounds on a finite horizon. The derivative bounds propagate by the quotient rule. Weighted tail control in x and Rellich compactness on every observation stratum give precompactness; the uniform positive bounds and the closed transition identities give closedness. Thus the reachable set, rather than the whole belief space, is compact and forward invariant. $\square$

1.2. The exact belief update and null observations. For $\pi \in \mathcal{K}$ define the predictive joint density

$$j_\vartheta^{a,\pi}(x', y) = g_\vartheta^a(x', y) \int r(x) m_\vartheta^a(x, x') m_0(dx)$$

and evidence

$$e_\vartheta^{a,\pi}(y) = \int j_\vartheta^{a,\pi}(x', y) m_0(dx').$$

For $e(y) > 0$, the posterior is

$$B_\vartheta^a(\pi, y)(dx') = \frac{j_\vartheta^{a,\pi}(x', y)}{e_\vartheta^{a,\pi}(y)} m_0(dx').$$

Adjoin one cemetery belief $\dagger$. On the Borel set $\{e = 0\}$ set the posterior equal to $\dagger$ and make $\dagger$ absorbing. This is a declared null-observation convention; the zero measure is not assigned a nonexistent projective direction.

Let

$$\mathbf{P}_{\vartheta}^a(\pi, d\pi') = \int \delta_{\mathbf{B}_{\vartheta}^a(\pi, y)}(d\pi') e_{\vartheta}^{a, \pi}(y) \nu(dy)$$

with the cemetery convention.

Theorem 1.2 (Feller belief kernel). *The space $\mathcal{Z} = \mathcal{K} \cup \{\dagger\}$ is compact metrizable and standard Borel. The map $(\vartheta, a, \pi, y) \mapsto \mathbf{B}_{\vartheta}^a(\pi, y)$ is Borel everywhere and continuous on every positive-evidence stratum. The belief transition $\mathbf{P}_{\vartheta}^a$ is Feller, and its restriction to the reachable positive-evidence set leaves $\mathcal{K}$ invariant.*

Proof. Weighted $L^1(m_0)$ metrizes the compact set $\mathcal{K}$ by Theorem 1.1; adding one isolated point preserves compact metrizability. The predictive density is jointly continuous in weighted L^1 by common domination. The evidence is therefore continuous. On $\{e > 0\}$, division is continuous and the elementary estimate

$$\left\| \frac{j}{e} - \frac{\tilde{j}}{\tilde{e}} \right\|_{L_W^1} \leq \frac{\|j - \tilde{j}\|_{L_W^1}}{e} + \frac{\|\tilde{j}\|_{L_W^1} |e - \tilde{e}|}{e\tilde{e}}$$

applies. On $\{e = 0\}$ the explicit cemetery definition is Borel because the set is closed.

For a continuous F on $\mathcal{Z}$, split the observation integral into $\{e \geq \delta\}$ and $\{e < \delta\}$. The first part is continuous by dominated convergence and the posterior estimate. The second has total predictive mass at most $\delta\nu(Y)$ on compact strata, with the tail controlled by the uniform observation moment. Let $\delta \downarrow 0$. This proves the Feller property. Forward invariance is part of the construction in Theorem 1.1. $\square$

1.3. Risk-sensitive partially observed control. Let $\mathcal{A}(\pi)$ be a nonempty compact set of relaxed actions with closed Borel graph, and let $c(\pi, a)$ and terminal reward $h(\pi)$ be bounded continuous functions on each energy/source shell. A policy is predictable with respect to the observation filtration. For risk parameter θ , set

$$V_n(\pi) = \sup_{\alpha} \frac{1}{\theta} \log \mathbb{E}_{\pi}^{\alpha} \exp \left\{ \theta \left[\sum_{k=0}^{n-1} c(\Pi_k, A_k) + h(\Pi_n) \right] \right\}.$$

Theorem 1.3 (Exact belief-state DPP and measurable selector). *The posterior process on $\mathcal{Z}$ is a controlled Markov process and*

$$V_{n+1}(\pi) = \frac{1}{\theta} \log \sup_{a \in \mathcal{A}(\pi)} \left[e^{\theta c(\pi, a)} \int e^{\theta V_n(\pi')} \mathbf{P}_{\vartheta}^a(\pi, d\pi') \right].$$

The supremum has a universally measurable maximizing selector, and a Borel selector exists when the control correspondence is continuous. The same

statement holds in continuous time for the B_4 relaxed kinetic controls on the compact action sublevels.

Proof. Prediction by M followed by observation by G and Bayes' formula shows that the conditional law of the next belief depends only on the present belief and action; the timing is therefore explicit. Conditional expectation and the tower property give the displayed recursion. The Feller theorem and continuity of the cost make the maximand upper semicontinuous in a relaxed action. Compactness gives attainment, and the measurable maximum theorem applies to the Borel graph.

For continuous time, use the B_4 path-space relaxed controls. Its compact action sublevels, lower semicontinuous cost, and positive transfer stability give the same measurable concatenation and selection properties. Exponential integrability follows from the fixed risk parameter and the action/energy bounds. $\square$

1.4. Finite coordinates on the reachable compact set. Choose a dense family (φ_j) in the unit ball of the weighted continuous hidden-state tests and define $T_m(\pi) = (\pi\varphi_1, \dots, \pi\varphi_m)$.

Proposition 1.4 (Quantitative finite-coordinate value approximation). *For every finite horizon n and $\varepsilon > 0$, there is m and a continuous reconstruction $R_m : T_m(\mathcal{K}) \rightarrow \mathcal{K}$ such that the policy obtained by applying an optimal belief policy to $R_m(T_m\pi)$ has value within ε uniformly over $\pi \in \mathcal{K}$. The required m is chosen on one forward-invariant reachable compact set.*

Proof. The coordinate maps separate points of the compact metric space $\mathcal{K}$. A finite subfamily therefore gives an embedding up to any prescribed metric tolerance; a partition of unity on its compact image gives a continuous approximate inverse. The belief kernel, cost, and exponential Bellman map are uniformly continuous on $\mathcal{K} \times \mathcal{A}$. Backward induction shows that one-step state errors are amplified by at most a finite horizon-dependent constant $C_{n,\theta}$, not by an unproved global contraction. Choose the reconstruction tolerance below $\varepsilon/C_{n,\theta}$. $\square$

1.5. Uniform triangular-array likelihood expansion. Let $P_{\vartheta}^{\varepsilon,\alpha}$ be an exact finite-volume observation experiment under a policy α in a compact regular strategy class. Its effective information scale is $a_\varepsilon \rightarrow \infty$ (collision returns for Sinai or μ_ε for hard spheres). Common domination yields a joint likelihood density $p_{\vartheta}^{\varepsilon,\alpha}$ on the complete observation path. Write

$$\dot{\ell}_{\vartheta}^{\varepsilon,\alpha} = \partial_{\vartheta} \log p_{\vartheta}^{\varepsilon,\alpha}, \quad \Delta_{\varepsilon}^{\alpha} = a_{\varepsilon}^{-1/2} \dot{\ell}_{\vartheta}^{\varepsilon,\alpha}.$$

Theorem 1.5 (Uniform LAQ and LAN). *On compact parameter, initial-belief, and strategy sets, for every bounded $h \in \mathbb{R}^p$,*

$$\log \frac{dP_{\vartheta+h/\sqrt{a_{\varepsilon}}}^{\varepsilon,\alpha}}{dP_{\vartheta}^{\varepsilon,\alpha}} = h^T \Delta_{\varepsilon}^{\alpha} - \frac{1}{2} h^T I_{\vartheta}^{\alpha} h + r_{\varepsilon}^{\alpha}(h),$$

where

$$\sup_{\alpha, |h| \leq H} P_{\vartheta}^{\varepsilon, \alpha}(|r_{\varepsilon}^{\alpha}(h)| > \eta) \rightarrow 0, \quad \Delta_{\varepsilon}^{\alpha} \implies N(0, I_{\vartheta}^{\alpha}),$$

and $cI \leq I_{\vartheta}^{\alpha} \leq CI$. The shifted experiments are contiguous.

Proof. Factor the likelihood into its predictive one-step densities generated by the exact filter. Taylor's formula with integral remainder gives, at $u = h/\sqrt{a_{\varepsilon}}$,

$$\ell_{\vartheta+u} - \ell_{\vartheta} = u^T \dot{\ell}_{\vartheta} + \frac{1}{2} u^T \ddot{\ell}_{\vartheta} u + \frac{1}{6} \partial_{\vartheta}^3 \ell_{\vartheta+tu} [u, u, u].$$

Theorem 1.1 bounds the third derivative relative to the predictive density on every stratum. The sum of its conditional expectations is $O(a_{\varepsilon})$, so the total third-order remainder is $O(|h|^3 a_{\varepsilon}^{-1/2})$ uniformly. The martingale identity for the score gives zero conditional mean. The conditional information averages converge uniformly by the A4/B4 spectral or action-compactness estimates, and the centered observed Hessian is negligible by the same martingale maximal bound.

The score is an additive functional of the exact history/filter process in the Sinai case and a density/contact functional in the hard-sphere case. A4's martingale approximation and B3's process CLT give its uniform Gaussian limit. The periodic/observability nondegeneracy estimates give the common information bounds. The likelihood expansion and Le Cam's first lemma then imply contiguity. $\square$

1.6. Uniform tests and Bernstein–von Mises. Assume a prior density π_0 continuous and positive near the true parameter and with exponentially integrable tails in the compact parameter chart. Let $\hat{\vartheta}_{\varepsilon}$ be any efficient one-step estimator obtained from the score and information.

Lemma 1.6 (Uniform posterior concentration). *For every $M_{\varepsilon} \rightarrow \infty$,*

$$\sup_{\alpha} P_{\vartheta}^{\varepsilon, \alpha} [\Pi^{\varepsilon}(\sqrt{a_{\varepsilon}}|\vartheta' - \vartheta| > M_{\varepsilon} \mid \mathcal{Y}) > \eta] \rightarrow 0.$$

Proof. Inside a fixed neighborhood, the uniform information lower bound and the LAN expansion yield likelihood-ratio tests with errors bounded by Ce^{-cM^2} for alternatives at local distance M . Cover each local annulus by $O(M^p)$ balls and take the maximum test. Outside that neighborhood, compact identifiability gives a positive Kullback–Leibler separation. The A2 or B2 source large-deviation bound, uniformly over the compact policy class, then gives exponentially consistent tests. Integrate the testing inequality against the prior and use its local thickness to lower-bound the posterior denominator. Summing annuli proves the claim. $\square$

Theorem 1.7 (Strategy-uniform Bernstein–von Mises theorem). *Uniformly over the compact regular strategy class, the posterior law of*

$$\sqrt{a_{\varepsilon}}(\vartheta' - \hat{\vartheta}_{\varepsilon})$$

converges in total variation, in $P_{\vartheta}^{\varepsilon, \alpha}$ -probability, to $N(0, (I_{\vartheta}^{\alpha})^{-1})$.

Proof. Lemma 1.6 reduces the posterior integral to a growing local ball. On that ball, Theorem 1.5 is uniform and the prior ratio $\pi_0(\vartheta + h/\sqrt{a})/\pi_0(\vartheta)$ tends uniformly to one on every fixed ball. Complete the square in $h^T \Delta - \frac{1}{2} h^T I h$. Gaussian tails and the exponentially small annular tests make the truncation error vanish as the local ball grows. The efficient one-step estimator differs from $\vartheta + I^{-1} \Delta / \sqrt{a}$ by $o_P(a^{-1/2})$ uniformly. Scheffé's lemma then gives total-variation convergence, with all estimates uniform in the declared strategy class. $\square$

Theorem 1.8 (Round-twenty-one C1 closure). *The information platform has a fully typed transition–observation model, a model-derived stratified dominated chart, an explicit standard-Borel null convention, a Feller belief kernel and risk-sensitive DPP, quantitative finite coordinates on the reachable compact set, a uniform triangular-array LAN theorem, and a separately proved strategy-uniform Bernstein–von Mises theorem.*

Proof. Combine Theorems 1.1, 1.2, 1.3, 1.5, and 1.7 with Proposition 1.4. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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